Md.Shalha Mucha Bhuyan

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Education

Chittagong University of Engineering and Technology (CUET), Bangladesh *2019 – 2024*
B.Sc in Electronics and Telecommunication Engineering (ETE)
- **GPA:** 3.20/4.0
- **Last Two Semesters GPA:** 3.40, 3.34

Research Interest

Machine Learning, Natural Language Processing, Computer Vision, and Cybersecurity

Experience

Associate Software Engineer *March 2005 – Present*
Cloudly.io, Dhaka Bangladesh
- Conducted applied research in NLP, LLMs, Agentic AI, Multimodal AI, and Computer Vision, focusing on LLM evaluation, fine-tuning for domain-specific tasks, and advancing application quality.
- Built and deployed LLM/ML applications by integrating FastAPI backends, containerizing with Docker, and automating ML workflows using GitHub Actions on cloud environments.
- Designed and implemented RAG pipelines with scalable vector databases to improve context retention and response accuracy of LLM applications.
- Currently working on the **NetAgent**, an agentic orchestration system that integrates multiple AI tools, performs real-time log analysis, and enhances operational intelligence and monitoring.

Undergraduate Thesis

Attention-Based Bengali Visual Question Answering: Enhancing Bengali Language Understanding of Multimodal AI Systems *June 2023 - June 2024*
Supervisor: Khaleda Akhter Sathi
- Developed a Bengali Visual Question Answering system that takes an image and a question in Bengali as input and produces contextually accurate answers by combining computer vision and natural language processing, designed for low-resource languages.
- Constructed a Bengali Visual Question-Answering (BVQA) dataset using zero-shot capabilities of LLMs to support research in low-resource language VQA.
- Proposed Multimodal Cross-Attention Network (MCRAN) leveraging pre-trained transformers and multi-head attention to fuse visual and textual features, outperforming baseline VQA methods.

Research Publications

BVQA: Connecting Language and Vision Through Attention for Open-Ended Question Answering, (IEEE Access, Q1, IF: 3.4) May 2025
Md. Shalha Mucha Bhuyan, Eftekhar Hossain, *et al.*
10.1109/ACCESS.2025.3540388

Advancing Bangla Sign Language Recognition Through Deep Learning Techniques July 2025
Fahmida Ahmed Ifty,**Md. Shalha Mucha Bhuyan**, *et al.*
10.1109/NCIM65934.2025.11159861

Technical Skills

Programming Languages:	Python, C, C++, JavaScript, SQL
Data Analysis:	Pandas, NumPy, Scikit-Learn, Matplotlib, Seaborn, SQL
Database Management:	PostgreSQL, Vector Databases (Pinecone, FAISS)
Machine Learning & AI:	TensorFlow, Keras, PyTorch, Hugging Face, Scikit-Learn, LLMs, RAG, LangChain, LangGraph, LangSmith, MCP Server
Backend & Deployment:	FastAPI, Django, Docker, Git, CI/CD (GitHub Actions), REST APIs
Web Development:	HTML, CSS, Bootstrap
Operating Systems:	Linux, Windows
Tools & Platforms:	VS Code, Jupyter Notebook, PyCharm, GitHub, GitLab

Projects

Health Assistant Chatbot *[GitHub]*

- Developed a medical domain chatbot using RAG to provide contextually accurate answers from user-uploaded medical documents.
- Built backend with FastAPI and integrated LLM with embeddings for semantic search and medical knowledge retrieval.
- Implemented scalable vector database (Pinecone) and deployed interactive frontend with Streamlit, improving response accuracy and user accessibility.
- **Tools Used:** LLM, FastAPI, RAG, Pinecone, Strenlit, Embedding

News & Research QA Assistant *[GitHub]*

- Built an AI-powered web application that enables users to query long-form content (news articles, research papers, academic documents) and receive concise summaries and answers.
- Implemented RAG pipelines with LangChain and FAISS to enhance document retrieval and improve response accuracy.
- **Tools Used:** Streamlit, Langchain, FAISS, LLM, RAG

Visual Question Answering (VQA) *[GitHub]*

- Developed a VQA application using ViLT (Vision-and-Language Transformer) from Hugging Face, enabling accurate answers to image-based queries through FastAPI backend and Streamlit UI.
- **Tools Used:** FastAPI, ViLT, RestAPI, Stremlit

Certifications

- **Fundamentals of MCP**, Hugging Face
- **MCP Course Unit 3: Automation in Production**, Hugging Face
- **Generative AI with Large Language Models**, Coursera
- **Introduction to Git and GitHub**,Google
- **Natural Language Processing with Sequence Models**, Coursera
- **TensorFlow Developer Professional Certificate** ,Coursera

Achievements

- **10th place in ML Competition,** CUET-ETE Day

Extracurricular activity

- Co-Finance Secretary, CUET Sports Club
- Contest Management Secretary,CUET ETE Day
- Champion, ETE Premier League (EPL), organized by the Department of ETE
- Attended Cultural Programme at ETE Day 2022, organized by the Department of ETE

Reference

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